

Interview VerseAI-An Agentic Intelligent Interview Assessment Ecosystem for Adaptive Skill Evaluation and Personalized Career Readiness

SOA Project Review-1

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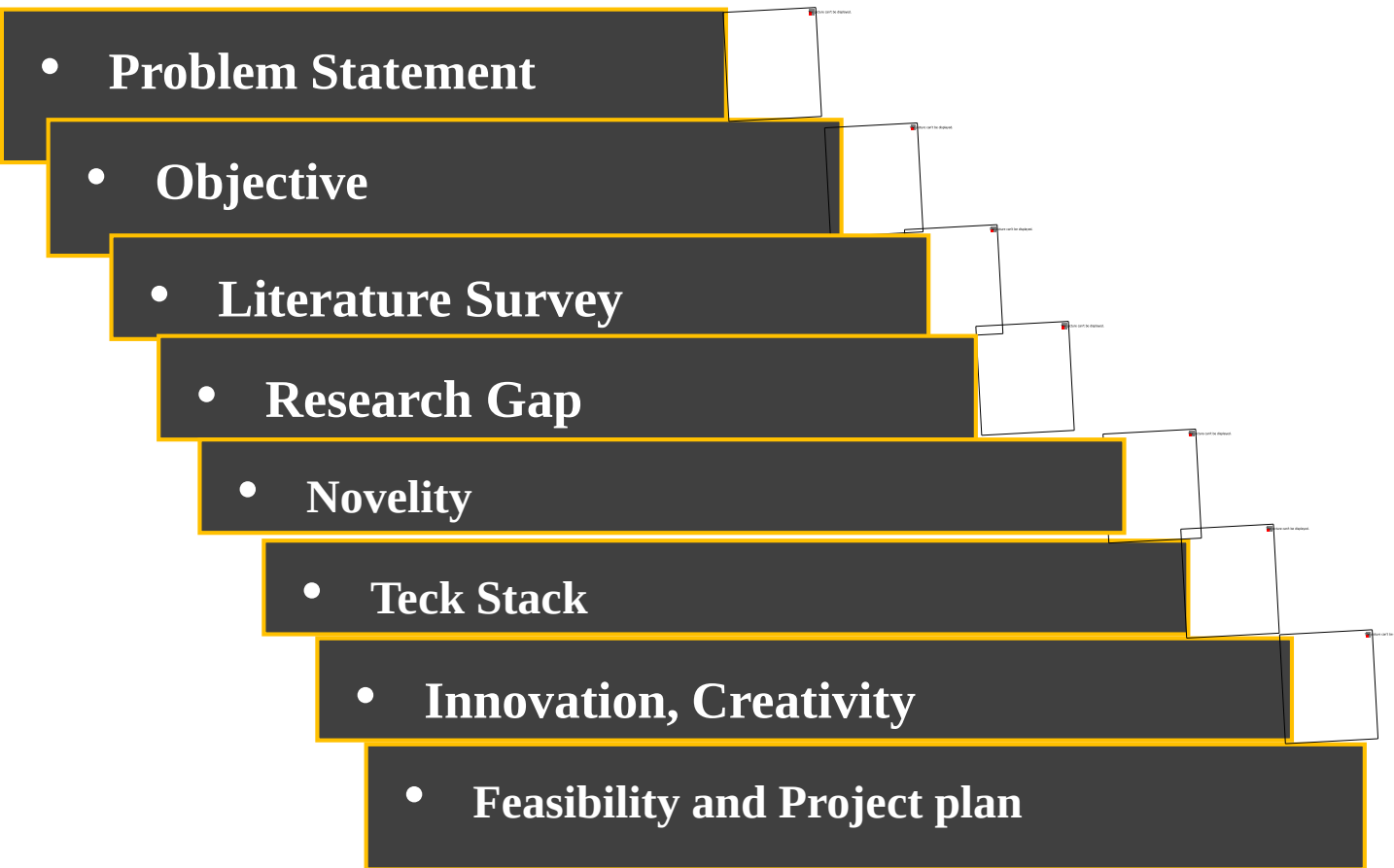
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Outline of Research Work

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- Problem Statement
 - Objective
 - Literature Survey
 - Research Gap
 - Novelty
 - Teck Stack
 - Innovation, Creativity
 - Feasibility and Project plan

Problem Statement

Problem Statement

Traditional interview processes often depend on **fixed questions, manual evaluation, and subjective judgment**, making it difficult to accurately assess the diverse skills of candidates. Existing online interview and assessment platforms generally focus on predefined question sets, technical tests, or basic scoring, with limited ability to adapt the interview according to a candidate's performance. Candidates may also receive scores without detailed information about their **strengths, weaknesses, skill gaps, and career readiness**. Therefore, there is a need for an intelligent interview assessment system that can dynamically generate and adjust questions, evaluate **technical, problem-solving, communication, and behavioural skills**, and provide personalized feedback. **InterviewVerseAI** addresses this problem by using an **Agentic AI-based adaptive interview approach** to transform interviews from a fixed test into an intelligent, personalized journey that helps candidates understand their capabilities and improve their career readiness.

Motivation

- **Adaptive Interview Experience:** Make interviews more dynamic by allowing AI to change the type and difficulty of questions based on the candidate's previous responses.
- **Comprehensive Skill Evaluation:** Evaluate not only technical knowledge but also **problem-solving, communication, and behavioural skills** to create a more complete candidate profile.
- **Personalized Feedback:** Provide candidates with clear information about their

Objectives

Objectives

1. To develop an **AI-powered intelligent interview assessment platform** for candidate evaluation.
2. To implement an **Agentic Interview System** capable of generating dynamic and context-aware questions.
3. To develop an **adaptive difficulty mechanism** that changes question complexity based on candidate performance.
4. To evaluate candidates across multiple dimensions such as:
 - Technical skills
 - Problem-solving
 - Communication
 - Behavioural skills
5. To use **NLP-based analysis** for understanding interview responses.
6. To identify candidate **strengths, weaknesses, and skill gaps**.
7. To generate **personalized feedback and improvement recommendations**.

Literature

Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
2020 – Yusuf & Lhaksmana	An Automated Interview Grading System in Talent Recruitment using SVM	Uses Support Vector Machine (SVM) for automated interview grading and candidate evaluation.	Demonstrates that machine learning can automate interview scoring and support talent recruitment.	Mainly focuses on automated grading; it does not provide adaptive questioning or personalized career guidance.
2024 – Chun et al.	A Study on the Design and Implementation of an AI Mock Interview System for Computer Science Interview Preparation Using LLM-based ChatGPT	Uses ChatGPT/LLM, NLP, Speech-to-Text and real-time feedback to analyze interview responses.	Demonstrates that LLMs can support realistic mock interviews and provide useful real-time feedback; 90% of surveyed users considered the feedback helpful.	Focuses mainly on interview preparation and feedback; future work includes expanding question types and industries.
2026 – Bershika & Golden Nancy	AI-Based Virtual Interviewer System Using NLP and Emotion Detection	Combines NLP, multimodal emotion recognition, Speech-to-Text, adaptive question generation and candidate-role matching.	Demonstrates a more comprehensive AI interviewer that dynamically adapts questions based on candidate performance and evaluates linguistic and	Still has scope for integrating personalized skill-gap analysis, career-readiness assessment and continuous candidate development into one ecosystem.

Research Gap



Existing Research / Approach	Research Gap Identified	Our Solution
1. Fixed and Non-Adaptive Interview Systems – Traditional interview platforms use a predefined set of questions for all candidates, regardless of their responses, skills, or difficulty level. These systems fail to adapt to individual performance, leading to a one-size-fits-all evaluation.	Most interview systems lack adaptability and real-time adjustment based on candidate responses. They do not personalize question flow, focus areas, or difficulty levels, resulting in less accurate assessment of true potential and skills.	Adaptive Interview Journey: InterviewVerseAI transforms interviews from a fixed test into an intelligent, adaptive journey. It continuously analyzes candidate responses and dynamically adjusts the next question, topic, and difficulty level, creating a personalized interview experience tailored to each candidate.
2. Single-Model Evaluation Approach – Conventional systems rely on a single evaluation model or interviewer, often focusing only on technical knowledge. This leads to a narrow assessment that overlooks communication, behavioral traits, and real-world problem-solving abilities.	There is a lack of comprehensive evaluation that covers multiple skill dimensions (technical, communication, behavior, planning, feedback) from diverse perspectives. Existing systems cannot simulate a real-world interview panel.	Agentic Multi-Skill Interview Assessment: InterviewVerseAI uses specialized AI agents (Interview, Technical, Communication, Behavioral, Planning, Feedback) that collaborate under an orchestration engine to evaluate candidates from multiple skill perspectives, providing a holistic and real-world interview experience.
3. Generic Feedback and No Career Guidance – Most platforms only provide overall scores or basic feedback. They do not analyze individual strengths, weaknesses, or skill gaps in depth, nor do they offer personalized recommendations for improvement or career readiness.	There is a gap in converting interview performance into actionable insights. Existing solutions do not provide personalized career readiness levels or guidance to help candidates improve and grow.	Personalized Career Readiness Engine: InterviewVerseAI analyzes overall performance to identify strengths, weaknesses, skill gaps, and career readiness levels. It delivers personalized feedback and improvement recommendations, transforming interview outcomes into a continuous career-development journey.

Novelty

1 - Adaptive Interview Journey

- InterviewVerseAI transforms the traditional fixed interview into an intelligent, adaptive interview journey. Instead of presenting the same predefined questions to every candidate, the system continuously analyses the candidate's responses and dynamically adjusts the next question, topic, and difficulty level according to individual performance. This creates a personalized interview experience that evolves with the candidate.

2 - Agentic Multi-Skill Interview Assessment

- Instead of relying on a single evaluation model, InterviewVerseAI employs specialized AI agents such as an Interview Agent, Technical Assessment Agent, Communication Agent, Behavioral Assessment Agent, and Feedback Agent. These agents collaborate under an Orchestration/Planning Agent to evaluate candidates from different perspectives and generate a comprehensive assessment covering technical knowledge, problem-solving, communication, and behavioural skills.

3 - Personalized Career Readiness Engine

- InterviewVerseAI goes beyond conventional interview scoring by analysing the candidate's overall performance to identify strengths, weaknesses, skill gaps, and career-readiness levels. The system provides personalized feedback and improvement recommendations based on the candidate's assessment results, helping transform interview performance into a continuous career-development journey.

Overall Novelty

InterviewVerseAI transforms interviews from a fixed test into an intelligent, adaptive journey toward personalized career readiness.

Tech Stack

Layer	Technology Stack	Purpose in InterviewVerseAI
Frontend	React	Candidate interview interface, question display, response submission, and performance dashboard
Backend	FastAPI	Handles AI services, interview logic, APIs, and communication between frontend and AI modules
Backend / Enterprise Services	Spring Boot	Manages application services, candidate data, assessment workflows, and backend integration
AI / LLM	GPT-5 API	Generates adaptive questions, follow-up questions, evaluates responses, and produces personalized feedback
Speech AI	Whisper	Converts candidate voice responses into text for interview analysis
Agentic AI	LangChain	Orchestrates the interview agent, manages context, and controls the adaptive question flow
Database	PostgreSQL	Stores candidate profiles, interview questions, responses, scores, skill

Innovation, Creativity

1. Agentic Interviewer

Existing: Fixed interviewer/question sequence.

Our Approach: AI agent dynamically decides what question should be asked next.

Benefit: Creates a personalized interview flow.

2. Adaptive Difficulty

Existing: Same difficulty for all candidates.

Our Approach: Difficulty changes according to candidate performance.

Benefit: Provides an appropriately challenging assessment.

3. Multi-Skill Evaluation

Existing: Technical tests are often separated from communication/behavioral evaluation.

Our Approach: Combines multiple skill dimensions.

Benefit: Provides a more comprehensive candidate profile.

4. Personalized Career Feedback

Existing: Assessment ends with a score.

Our Approach: Converts assessment results into personalized improvement recommendations.

Benefit: Helps candidates prepare for future interviews and career opportunities.

Feasibility

1. Technical Feasibility

The system is technically feasible using existing **NLP, LLM, machine-learning, speech-to-text, and web technologies**.

The adaptive agent can be implemented using existing AI/LLM frameworks, while candidate responses can be processed using NLP models.

2. Economic Feasibility

The prototype can be developed using:

- Open-source AI/ML libraries
- Public datasets
- Open-source development frameworks
- Existing pretrained models

This reduces development cost.

3. Operational Feasibility

The system provides a simple workflow: **Candidate → Interview → AI Assessment → Feedback → Career Report**

Candidates can interact with the system without requiring specialized technical knowledge.

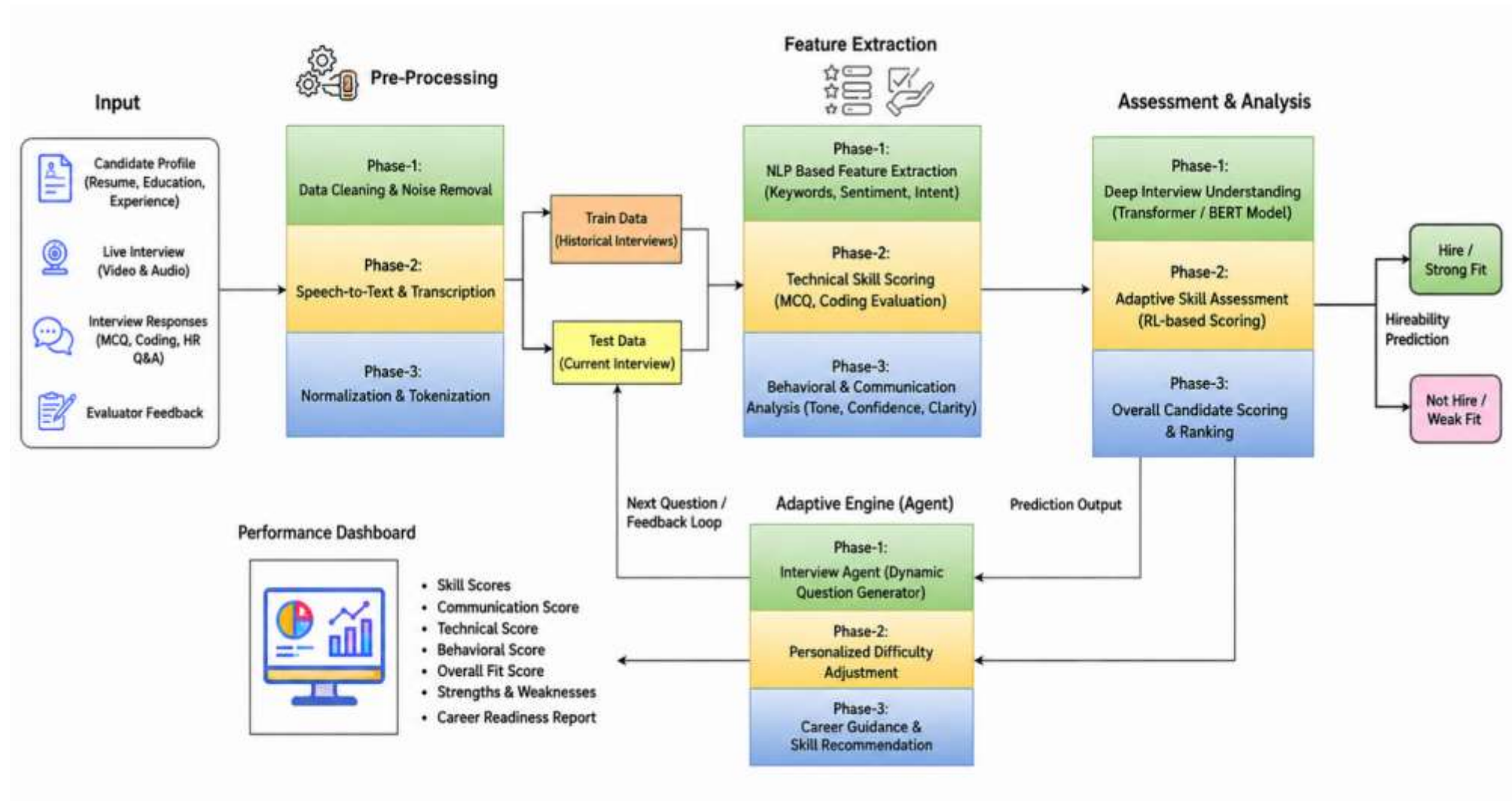
4. Integration Feasibility

Different components can be integrated through APIs: **Frontend ↔ Backend ↔ AI Models ↔ Database ↔ Dashboard**

Overall Feasibility

The project is feasible because it uses **existing AI technologies in a new integrated architecture**. The major contribution is the combination of **agentic interviewing, adaptive**

Project Architecture/Project Overview:



PROJECT PLAN

Phase	Activity	Expected Output
1	Literature Study & Requirements	Research gap & requirements
2	Dataset Preparation	Interview dataset
3	Pre-processing & Feature Extraction	Processed interview features
4	AI Assessment	Skill scores & analysis
5	Adaptive Agent	Dynamic questions & difficulty
6	Feedback & Career Analysis	Personalized recommendations
7	Dashboard & Integration	Complete working prototype
8	Testing & Documentation	Evaluation results & final demo